FINAL TERM – ACTIVITY 1

Inline, Embedded, and External Styles | CSS Selectors

LEARNING OBJECTIVES

By the end of this activity, you should be able to:

- Apply CSS using inline, embedded, and external styles, and explain when each one is appropriate.
- Predict which rule wins when several rules target the same element.
- Use universal, element, class, ID, and grouping selectors.
- Use descendant, child, adjacent sibling, and general sibling combinators.
- Use pseudo-class selectors and attribute selectors.

WHAT YOU NEED

- A computer with a code editor (VS Code, Notepad++, or similar)
- A web browser (Chrome, Edge, or Firefox)
- Basic knowledge of HTML tags and page structure

Part 1 Set Up and Build the Starter Page

You will start with a page that has no styling at all. This is the page you will style in every part that follows.

> **How to work** Type the code yourself instead of copying it. After every step, save the file and refresh the browser. Where a step says Predict, write your guess first, then test it.

Step 1 Create a folder named “css-lab” on the Desktop or in your Documents folder.

Step 2 Open the folder in your code editor and create a file named “index.html”.

Step 3 Type the code below. Watch the `id` and `class` attributes closely, because later parts depend on them.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Campus Club Fair</title>
</head>
<body>

  <header id="top">
    <h1>Campus Club Fair 2026</h1>
    <p>Find your people. Join a club.</p>
  </header>

  <nav>
    <a href="#clubs">Clubs</a>
    <a href="#schedule">Schedule</a>
    <a href="#signup">Sign Up</a>
    <a href="https://developer.mozilla.org/" target="_blank">CSS Help</a>
  </nav>

  <main>
    <section id="clubs">
      <h2>Featured Clubs</h2>
      <p>Three clubs are opening their doors this semester.</p>
      <p class="note">Membership is free for all students.</p>

      <div class="card">
        <h3>Code Guild</h3>
        <p>Weekly coding sessions and hackathons.</p>
        <p>Meets every Tuesday, 4:00 PM.</p>
      </div>

      <div class="card featured">
        <h3>Robotics Circle</h3>
        <p>Build and program robots as a team.</p>
        <p>Meets every Wednesday, 3:00 PM.</p>
      </div>

      <div class="card">
        <h3>Design Lab</h3>
        <p>Posters, layouts, and user interface design.</p>
        <p>Meets every Friday, 1:00 PM.</p>
      </div>
    </section>
```

index.html (continued)

```
    <section id="schedule">
      <h2>Schedule</h2>
      <p>All events happen at the covered court.</p>
      <ul>
        <li>8:00 AM - Opening program</li>
        <li>9:00 AM - Club booths open</li>
        <li>11:00 AM - Robotics demo</li>
        <li>1:00 PM - Design showcase</li>
        <li>3:00 PM - Closing and raffle</li>
      </ul>
    </section>

    <section id="signup">
      <h2>Sign Up</h2>
      <form>
        <label for="fullname">Full name</label>
        <input type="text" id="fullname" name="fullname">
        <label for="email">Email</label>
        <input type="email" id="email" name="email">
        <input type="submit" value="Join">
      </form>
    </section>
  </main>

  <footer>
    <p>Campus Club Fair 2026. Made by (your name).</p>
  </footer>

</body>
</html>
```

Step 4 Replace `(your name)` in the footer with your own name. Save the file.

Step 5 Open `index.html` in the browser. You should see plain black text on a white background, with every section stacked from top to bottom.

Part 2 Inline CSS

Inline CSS is written inside the opening tag of one element, using the `style` attribute. It affects that one element only.

> **Pattern**
> `<tag style="property: value; property: value;">`

Step 1 Find the `h1` in the header and add a `style` attribute to it.

> **index.html**
> `<h1 style="color: darkblue; text-align: center;">Campus Club Fair 2026</h1>`

Step 2 Style the paragraph under it the same way.

> **index.html**
> `<p style="text-align: center; font-style: italic;">Find your people. Join a club.</p>`

Step 3 Save and refresh. The title should be dark blue and centered, and the tagline should be centered and italic.

Step 4 Now make all three club names orange. Add this `style` attribute to each of the three `h3` elements, one at a time.

> **index.html**
> `<h3 style="color: darkorange;">Code Guild</h3>`

Step 5 The club officers changed their minds. Change all three club names from `darkorange` to `teal`. Count how many places you had to edit.

> **Notice** Every element carries its own copy of the style. Nothing is shared.

Part 3 Embedded CSS

Embedded CSS (also called internal CSS) is written inside a `<style>` element in the `<head>` of the page. One rule can style many elements on that page.

Pattern

```
selector {
  property: value;
}
```

Step 1 Remove the `style` attribute from all three `h3` elements. Leave the inline styles on the `h1` and the tagline for now.

Step 2 Inside `<head>`, below the `<title>` line, add a `<style>` block.

index.html

```
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 40px;
  }

  h3 {
    color: teal;
  }
</style>
```

Step 3 Save and refresh. All three club names should be teal again, and the whole page should use the Arial font.

Step 4 Change the `h3` color to `darkorange`, refresh, then change it back to `teal`. Count your edits this time.

Step 5 Predict: you are about to add the rule below inside the `<style>` block. Will the main title turn green? Write your prediction in Checkpoint 3, question 3, then add the rule and test it.

index.html (inside <style>)

```
h1 {
  color: green;
}
```

> **Keep it** Do not delete the green `h1` rule yet. You will need it in Part 4.

Part 4 External CSS and the Cascade

25 min

External CSS is written in a separate `.css` file. Any page can use it by linking to the file, so one stylesheet can style an entire website.

Step 1 In the `css-lab` folder, create a new file named `styles.css`. It must sit beside `index.html`.

Step 2 Cut the `body` rule and the `h3` rule out of the `<style>` block and paste them into `styles.css`. Leave the green `h1` rule where it is.

styles.css

```
body {
  font-family: Arial, sans-serif;
  margin: 40px;
}

h3 {
  color: teal;
}
```

> **Careful** A `.css` file contains CSS rules only. Do not type `<style>` tags inside it.

Step 3 Connect the stylesheet. In `index.html`, add this line inside `<head>`, above the `<style>` block.

index.html

```
<link rel="stylesheet" href="styles.css">
```

Step 4 Save both files and refresh. The page should look exactly as it did at the end of Part 3. If the font and the teal color disappeared, check the spelling of the file name and confirm that both files are in the same folder.

Experiment: which style wins?

Step 5 Add one more rule at the bottom of `styles.css`.

styles.css

```
h1 {
  color: crimson;
}
```

The `h1` now receives three different colors: `darkblue` from the inline style, `green` from the embedded rule, and `crimson` from the external file. Run the four trials below in order. For each one, write your prediction first, then make the change, refresh, and record what you see.

Step 6 Clean up. Remove the inline `style` attribute from the tagline paragraph too. In `styles.css`, change the `h1` color from `crimson` to `darkblue`. From here on, all your CSS goes in `styles.css`.

Part 5 Basic Selectors

A selector is the part of a CSS rule that says which elements the rule applies to. Add every rule in this part to the bottom of `styles.css`, in the order given.

Universal selector: *

Step 1 The universal selector matches every element. Add this temporary rule to see just how many elements are on your page.

styles.css

```
* {
  border: 1px solid red;
}
```

Step 2 Look at the result, then replace the rule with a more useful one.

styles.css

```
* {
  box-sizing: border-box;
}
```

Element selector: tagname

Step 3 An element selector matches every element with that tag name.

styles.css

```
h2 {
  color: darkblue;
  border-bottom: 2px solid teal;
  padding-bottom: 4px;
}
```

Class selector: .classname

Step 4 A class selector starts with a dot and matches every element that has that class. A class can be reused on many elements.

styles.css

```
.card {
  background-color: #f4f6f8;
  border: 1px solid #cccccc;
  padding: 16px;
  margin-bottom: 12px;
}

.featured {
  border: 3px solid darkorange;
}

.note {
  color: gray;
  font-size: 14px;
}
```

Look closely The Robotics Circle card has both classes, so it receives the `.card` rule and the `.featured` rule at the same time.

ID selector: #idname

Step 5 An ID selector starts with a hash sign and matches the single element with that `id`. An `id` value must appear only once per page.

styles.css

```
#top {
  background-color: darkblue;
  color: white;
  padding: 24px;
  text-align: center;
}
```

Step 6 Refresh and look at the header. The tagline is white, but the main title seems to have disappeared. Before reading further, try to work out why.

> **What happened** The title is still there. Your earlier `h1` rule sets its color to `darkblue`, which is now the same as the background. The white color from `#top` is only inherited by the `h1`, and a rule that targets an element directly always beats an inherited value.

Step 7 Fix it by changing the color in your existing `h1` rule from `darkblue` to `white`.

Grouping selector: A, B

Step 8 A comma lets several selectors share one rule.

styles.css

```
h1, h2, h3 {
  font-family: Georgia, serif;
}
```

Experiment: class against element

Step 9 Predict: you are about to add the rule below at the very bottom of the file. It comes after the `.note` rule. Will the gray note turn black? Write your prediction in Checkpoint 5, question 4, then test it.

styles.css

```
p {
  color: black;
}
```

Step 10 Record the result, then delete the `p` rule you just added. It was only for this experiment, and it would also turn the white tagline in the header black.

Part 6 Combinator Selectors

Combinators join two selectors and describe how the elements are related in the HTML structure. Keep adding rules to the bottom of `styles.css`.

Combinator	Written as	Selects
Descendant	`A B`	Every `B` anywhere inside `A`
Child	`A > B`	Every `B` that is directly inside `A`, one level down
Adjacent sibling	`A + B`	The `B` that comes immediately after `A`, with the same parent
General sibling	`A ~ B`	Every `B` that comes after `A`, with the same parent

Descendant: A B

Step 1 Style the tagline without giving it a class. It is the only paragraph inside the header.

styles.css
```
header p {
  font-style: italic;
}
```

Step 2 Now mark every paragraph that is inside a `section`.

styles.css
```
section p {
  border-left: 4px solid teal;
  padding-left: 10px;
}
```

Step 3 Refresh and count the paragraphs that have a teal bar on the left. Write the number in Checkpoint 6, question 1.

Child: A > B

Step 4 Change the selector in the rule from Step 2. Replace the space with a `>` sign.

styles.css
```
section > p {
  border-left: 4px solid teal;
  padding-left: 10px;
}
```

Step 5 Refresh and count again. Look at the HTML to see what the remaining paragraphs have in common.

Adjacent sibling: A + B

Step 6 Make the first line under each club name stand out.

styles.css
```
h3 + p {
  font-weight: bold;
}
```

General sibling: A ~ B

Step 7 Indent every paragraph that follows a club name.

styles.css

```
h3 ~ p {
  margin-left: 20px;
}
```

Step 8 Refresh. In each card, both paragraphs should be indented, but only the first one should be bold.

Part 7 Pseudo-class and Attribute Selectors

Pseudo-classes

A pseudo-class starts with a colon. It selects an element because of its state (such as being pointed at) or its position among its siblings, not because of a tag, class, or id.

Step 1 Style the navigation links, then give them a different look while the mouse pointer is over them.

styles.css

```
nav {
  margin: 16px 0;
}

nav a {
  color: darkblue;
  text-decoration: none;
  margin-right: 16px;
}

nav a:hover {
  color: darkorange;
  text-decoration: underline;
}
```

Step 2 Refresh and move the pointer over each link.

Step 3 Style the schedule list by position.

styles.css

```
li {
  padding: 6px;
}

li:first-child {
  font-weight: bold;
}

li:last-child {
  color: crimson;
}

li:nth-child(even) {
  background-color: #e3ecf5;
}
```

Step 4 Refresh. Note which rows are shaded. Then change `even` to `odd`, observe, and change it back.

Step 5 Highlight a form field while the user is typing in it.

styles.css

```
input:focus {
  outline: 3px solid darkorange;
}
```

Step 6 Refresh, click inside each text box, and press the Tab key to move between them.

Attribute selectors

An attribute selector is written in square brackets. It selects elements by an attribute or by the value of an attribute.

Written as	Selects
`[target]`	Elements that have a `target` attribute, with any value
`[type="text"]`	Elements whose `type` is exactly `text`
`[href^="https"]`	Elements whose `href` starts with `https`

Step 7 Lay out the form, then style each kind of input differently even though all three use the same `input` tag.

styles.css

```
label {
  display: block;
  margin-top: 10px;
}

input {
  display: block;
  padding: 8px;
  margin-top: 4px;
}

input[type="text"] {
  border: 2px solid teal;
}

input[type="email"] {
  border: 2px solid darkorange;
}

input[type="submit"] {
  margin-top: 14px;
  padding: 8px 24px;
  background-color: darkblue;
  color: white;
  border: none;
}
```

Step 8 Mark the link that leaves the website so visitors can tell it apart from the links that jump within the page.

styles.css

```
a[href^="https"] {
  font-style: italic;
}

a[target="_blank"] {
  border-bottom: 1px dashed darkblue;
}
```

Step 9 Refresh. Only the CSS Help link should be italic with a dashed underline.

Part 8 Final Check and Submission

Go through the list and tick each item once you have confirmed it.

Done	Requirement
	`index.html` has no `style` attributes and no `<style>` block.
	`index.html` links to `styles.css` with a `<link>` tag inside `<head>`.
	The header has a dark blue background with a visible white title and an italic tagline.
	All three club cards are styled, and the Robotics Circle card has an orange border.
	Navigation links change color when the pointer is over them.
	The schedule list has a bold first item, a crimson last item, and shaded even rows.
	The text, email, and submit inputs each look different, and a focused input shows an orange outline.
	Your name appears in the footer.
	All checkpoints in this handout are answered.

Selector quick reference

Selector	Example	What it selects
Universal	`*`	Every element
Element	`h2`	Every `h2` element
Class	`.card`	Every element with `class="card"`
ID	`#top`	The one element with `id="top"`
Grouping	`h1, h2, h3`	All `h1`, `h2`, and `h3` elements
Descendant	`section p`	Paragraphs anywhere inside a section
Child	`section > p`	Paragraphs directly inside a section
Adjacent sibling	`h3 + p`	The paragraph right after an `h3`
General sibling	`h3 ~ p`	All paragraphs after an `h3`, same parent
Pseudo-class	`a:hover`	A link while the pointer is over it
Attribute	`input[type="text"]`	Inputs whose type is `text`

Submission

Compress the `css-lab` folder into a ZIP file named `Surname_CSSLab.zip` and submit it together with this handout, following your instructor's directions.